

► REPRODUCTIVE—PATHOLOGY

Sex chromosome disorders

Aneuploidy most commonly due to meiotic nondisjunction.

Klinefelter syndrome**A**

Male, 47,XXY.
Small, firm testes; infertility (azoospermia); tall stature with eunuchoid proportions (delayed epiphyseal closure → ↑ long bone length); gynecomastia **A**; female hair distribution. May present with developmental delay. Presence of inactivated X chromosome (Barr body). Common cause of hypogonadism seen in infertility workup. ↑ risk of breast cancer.

Dysgenesis of seminiferous tubules
→ ↓ inhibin B → ↑ FSH.
Abnormal Leydig cell function → ↓ testosterone
→ ↑ LH.

Turner syndrome**B**

Female, 45,XO.
Short stature (preventable with GH therapy), ovarian dysgenesis (streak ovary), broad chest with widely spaced nipples, bicuspid aortic valve, coarctation of the aorta (femoral < brachial pulse), lymphatic defects (result in webbed neck **B** or cystic hygroma; lymphedema in feet, hands), horseshoe kidney, high-arched palate, shortened 4th metacarpals. Most common cause of 1° amenorrhea. No Barr body.

Menopause before menarche.
↓ estrogen leads to ↑ LH, FSH.
Sex chromosome (X, or rarely Y) loss often due to nondisjunction during meiosis or mitosis.
Meiosis errors usually occur in paternal gametes
→ sperm missing the sex chromosome.
Mitosis errors occur after zygote formation → loss of sex chromosome in some but not all cells
→ mosaic karyotype (eg, 45,X/46,XX).
(45,X/46,XY) mosaicism associated with ↑ risk for gonadoblastoma.
Pregnancy is possible in some cases (IVF, exogenous estradiol-17β and progesterone).

Double Y males

47,XXY.
Phenotypically normal (usually undiagnosed), very tall. Normal fertility. May be associated with severe acne, learning disability, autism spectrum disorders.

Other disorders of sex development

Formerly called intersex states. Discrepancy between phenotypic sex (external genitalia, influenced by hormonal levels) and gonadal sex (testes vs ovaries, corresponds with Y chromosome).

46,XX DSD

Ovaries present, but external genitalia are virilized or atypical. Most commonly due to congenital adrenal hyperplasia (excessive exposure to androgens early in development).

46,XY DSD

Testes present, but external genitalia are feminized or atypical. Most commonly due to androgen insensitivity syndrome (defect in androgen receptor).

Ovotesticular DSD

46,XX > 46,XY. Both ovarian and testicular tissue present (ovotestis); atypical genitalia.

Diagnosing disorders by sex hormones	Testosterone	LH	Diagnosis
	↑	↑	Androgen insensitivity syndrome
	↑	↓	Testosterone-secreting tumor, exogenous androgenic steroids
	↓	↑	Hypergonadotropic (1°) hypogonadism
	↓	↓	Hypogonadotropic (2°) hypogonadism

Diagnosing disorders by physical characteristics	Uterus	Breasts	Diagnosis
	⊕	⊖	Hypergonadotropic (1°) hypogonadism in genotypic female Hypogonadotropic (2°) hypogonadism in genotypic female
	⊖	⊕	Müllerian agenesis in genotypic female Androgen insensitivity syndrome in genotypic male

Aromatase deficiency	Inability to synthesize endogenous estrogens. Autosomal recessive. During fetal life, DHEA produced by fetal adrenal glands cannot be converted to estrogen by the placenta and is converted to testosterone peripherally → virilization of both female infant (atypical genitalia) and mother (acne, hirsutism; fetal androgens can cross placenta).
-----------------------------	---

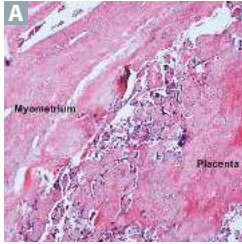
Androgen insensitivity syndrome	Defect in androgen receptor resulting in female-appearing genetic male (46,XY DSD); breast development and female external genitalia with scant axillary and pubic hair, rudimentary vagina; uterus and fallopian tubes absent due to persistence of anti-Müllerian hormone from testes. Patients develop normal functioning testes (often found in labia majora; surgically removed to prevent malignancy). ↑ testosterone, estrogen, LH (vs sex chromosome disorders).
--	--

5α-reductase deficiency	Autosomal recessive; sex limited to genetic males (46,XY DSD). Inability to convert testosterone to DHT. Atypical genitalia until puberty, when ↑ testosterone causes masculinization/↑ growth of external genitalia. Testosterone/estrogen levels are normal; LH is normal or ↑. Internal genitalia are normal.
--------------------------------	--

Kallmann syndrome	Failure to complete puberty; a form of hypogonadotropic hypogonadism. Defective migration of neurons and subsequent failure of olfactory bulbs to develop → ↓ synthesis of GnRH in the hypothalamus; hyposmia/anosmia; ↓ GnRH, FSH, LH, testosterone. Infertility (low sperm count in males; amenorrhea in females).
--------------------------	--

Placental disorders

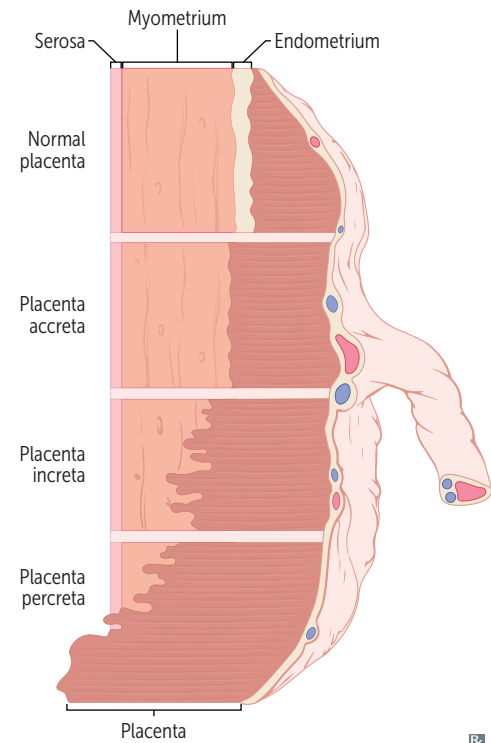
Placenta accreta spectrum



Abnormal invasion of trophoblastic tissue into uterine wall **A**. Risk factors: prior C-section or other uterine surgery (areas of uterine scarring impair normal decidualization), placenta previa, ↑ maternal age, multiparity. Three types depending on depth of trophoblast invasion:

- **Placenta accreta**—attaches to myometrium (instead of overlying decidua basalis) without invading it. Most common type.
- **Placenta increta**—partially invades **into** myometrium.
- **Placenta percreta**—completely invades (“**perforates**”) through myometrium and serosa, sometimes extending into adjacent organs (eg, bladder → hematuria).

Presents with difficulty separating placenta from uterus after fetal delivery and severe postpartum hemorrhage upon attempted manual removal of placenta (often extracted in pieces). Treatment: hysterectomy.



Placenta previa

Attachment of placenta over internal cervical os (a “**preview**” of the placenta is visible through cervix). Risk factors: prior C-section, multiparity.

Presents with painless vaginal bleeding in third trimester.

Low-lying placenta—located < 2 cm from, but not covering, the internal cervical os.

Vasa previa

Fetal vessels run over, or < 2 cm from, the internal cervical os. Risk factors: velamentous insertion of umbilical cord (inserts in chorioamniotic membrane rather than placenta → fetal vessels travel to placenta unprotected by Wharton jelly), bilobed or succenturiate placenta.

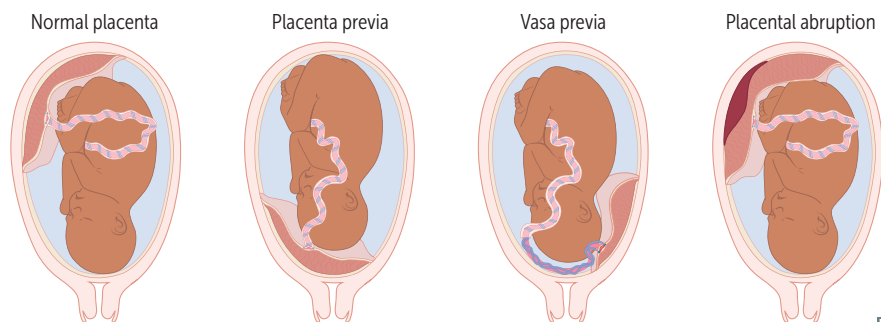
Presents with painless vaginal bleeding (fetal blood from injured vessels) upon rupture of membranes accompanied by fetal heart rate abnormalities (eg, bradycardia). May lead to fetal death from exsanguination.

Placental abruption

Also called abruptio placentae. Premature separation of placenta from uterus prior to fetal delivery.

Risk factors: maternal hypertension, preeclampsia, smoking, cocaine use, abdominal trauma.

Presents with **abrupt**, painful vaginal bleeding in third trimester; can lead to maternal hypovolemic shock (due to hemorrhage) and DIC (due to release of tissue factor from injured placenta), fetal distress (eg, hypoxia). May be life threatening for both mother and fetus.



Uterine rupture

Full-thickness disruption of uterine wall. Risk factors: prior C-section (usually occurs during labor in a subsequent pregnancy), abdominal trauma. Presents with painful vaginal bleeding, fetal heart rate abnormalities (eg, bradycardia), easily palpable fetal parts, loss of fetal station. May be life threatening for both mother and fetus.

Postpartum hemorrhage

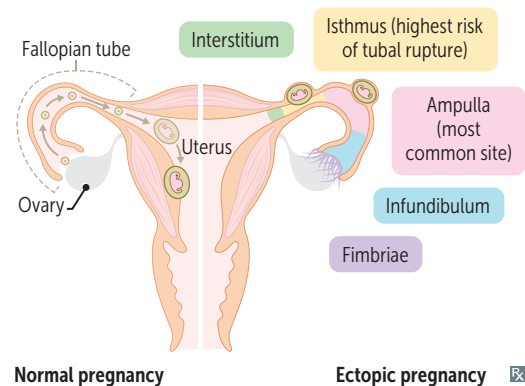
Greater-than-expected blood loss after delivery. Leading cause of maternal mortality worldwide. Etiology (**4 T's**): **T**one (uterine atony → soft, boggy uterus; most common), **T**rauma (eg, lacerations, incisions, uterine rupture), **T**issue (retained products of conception), **T**hrombin (coagulopathy). Treatment: uterine massage, oxytocin. If refractory, surgical ligation of uterine or internal iliac arteries (fertility is preserved since ovarian arteries provide collateral circulation).

Ectopic pregnancy

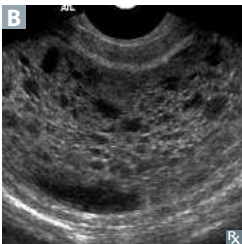
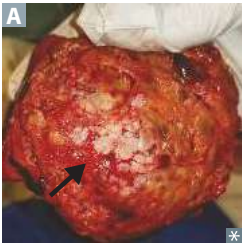
Implantation of fertilized ovum in a site other than the uterus, most often in ampulla of fallopian tube **A**. Risk factors: tubal pathologies (eg, scarring from salpingitis [PID] or surgery), previous ectopic pregnancy, IUD, IVF.

Presents with first-trimester bleeding and/or lower abdominal pain. Often clinically mistaken for appendicitis. Suspect in patients with history of amenorrhea, lower-than-expected rise in hCG based on dates. Confirm with ultrasound, which may show extraovarian adnexal mass.

Treatment: methotrexate, surgery.



Hydatidiform mole

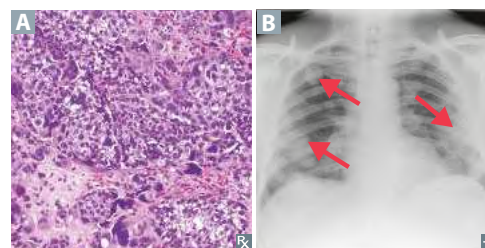


Cystic swelling of chorionic villi and proliferation of chorionic epithelium (only trophoblast). Presents with vaginal bleeding, emesis, uterine enlargement more than expected, pelvic pressure/pain. Associated with hCG-mediated sequelae: hyperthyroidism, theca lutein cysts, hyperemesis gravidarum, early preeclampsia (before 20 weeks of gestation). Treatment: dilation and curettage +/- methotrexate. Monitor hCG.

	Complete mole	Partial mole
KARYOTYPE	46,XX (most common); 46,XY	69,XXX; 69,XXY; 69,XYY
COMPONENTS	Most commonly enucleated egg + single sperm (subsequently duplicates paternal DNA)	2 sperm + 1 egg
HISTOLOGY	Hydropic villi, circumferential and diffuse trophoblastic proliferation	Only some villi are hydropic, focal/minimal trophoblastic proliferation
FETAL PARTS	No	Yes (partial = fetal parts)
STAINING FOR P57 PROTEIN	⊖ (paternally imprinted) C omplete m ole is c omplete m ale	⊕ (maternally expressed) P artial mole is P 57 p ositive
UTERINE SIZE	↑	—
hCG	↑↑↑↑	↑
IMAGING	“Honeycombed” uterus or “clusters of grapes” A , “snowstorm” B on ultrasound	Fetal parts
RISK OF INVASIVE MOLE	15–20%	< 5%
RISK OF CHORIOCARCINOMA	2%	Rare

Choriocarcinoma

Rare malignancy of trophoblastic tissue **A** (cytotrophoblasts, syncytiotrophoblasts), without chorionic villi present. Most commonly occurs after an abnormal pregnancy (eg, hydatidiform mole, abortion); can occur nongestationally in gonads. Presents with abnormal uterine bleeding, hCG-mediated sequelae, dyspnea, hemoptysis. Hematogenous spread to lungs → “cannonball” metastases **B**. Treatment: methotrexate.

**Hypertension in pregnancy****Gestational hypertension**

BP > 140/90 mm Hg after 20 weeks of gestation. No preexisting hypertension. No proteinuria or end-organ damage. Hypertension prior to 20 weeks of gestation suggests chronic hypertension. Treatment: antihypertensives (**H**ydralazine, **α**-**m**ethyldopa, **l**abetalol, **n**ifedipine), deliver at 37–39 weeks. **H**ypertensive **m**oms **l**ove **n**ifedipine.

Preeclampsia

New-onset hypertension with either proteinuria or end-organ dysfunction after 20 weeks of gestation (onset of preeclampsia < 20 weeks of gestation may suggest molar pregnancy). Caused by abnormal placental spiral arteries → endothelial dysfunction, vasoconstriction, ischemia. Risk factors: history of preeclampsia, multifetal gestation, nulliparity, chronic hypertension, diabetes, chronic kidney disease, autoimmune disorders (eg, antiphospholipid syndrome), obesity. Complications: placental abruption, coagulopathy, renal failure, pulmonary edema, uteroplacental insufficiency; may lead to eclampsia and/or HELLP syndrome. Treatment: antihypertensives, IV magnesium sulfate (to prevent seizure); definitive is delivery. Prophylaxis: aspirin.

Eclampsia

Preeclampsia with seizures. Death due to stroke, intracranial hemorrhage, ARDS. Treatment: IV magnesium sulfate, antihypertensives, immediate delivery.

HELLP syndrome

Preeclampsia with thrombotic microangiopathy of the liver. **H**emolysis, **E**levated **L**iver enzymes, **L**ow **P**latelets. May occur in the absence of hypertension and proteinuria. Blood smear shows schistocytes. Can lead to hepatic subcapsular hematomas (rupture → severe hypotension) and DIC (due to release of tissue factor from injured placenta). Treatment: immediate delivery.

Supine hypotensive syndrome

Also called aortocaval compression syndrome. Seen at > 20 weeks of gestation. Supine position → compression of abdominal aorta and IVC by gravid uterus → ↓ placental perfusion (can lead to pregnancy loss) and ↓ venous return (hypotension). Relieved by left lateral decubitus position.

Gynecologic tumor epidemiology

Incidence (US)—endometrial > ovarian > cervical; cervical cancer is more common worldwide due to lack of screening or HPV vaccination.

Prognosis: **Cervical** (**best** prognosis, diagnosed < 45 years old) > **Endometrial** (middle-aged, about 55 years old) > **Ovarian** (**worst** prognosis, > 65 years).

CEOs often go from **best** to **worst** as they get older.

Vulvar pathology**Non-neoplastic****Bartholin cyst and abscess**

Due to blockage of Bartholin gland duct causing accumulation of gland fluid. May lead to abscess 2° to obstruction and inflammation **A**. Usually in reproductive-age females.

Lichen sclerosus

Chronic, progressive inflammatory disease characterized by porcelain-white plaques **B** that can be hemorrhagic, eroded, or ulcerated. May extend to anus producing figure-eight appearance. ↑ incidence in prepubertal and peri-/postmenopausal females. Presents with intense pruritus, dyspareunia, dysuria, dyschezia. Benign, but slightly ↑ risk for SCC.

Lichen simplex chronicus

Hyperplasia of vulvar squamous epithelium. Presents with leathery, thick vulvar skin with enhanced skin markings due to chronic rubbing or scratching. Benign, no risk of SCC.

Neoplastic**Vulvar carcinoma**

Carcinoma from squamous epithelial lining of vulva **C**. Usually seen in postmenopausal females. Presents with leukoplakia, biopsy often required to distinguish carcinoma from other causes. HPV-related vulvar carcinoma—associated with high-risk HPV types 16, 18. Non-HPV vulvar carcinoma—usually from long-standing lichen sclerosus.

Extramammary Paget disease

Intraepithelial adenocarcinoma. Carcinoma in situ, low risk of underlying carcinoma (vs Paget disease of the breast, which is always associated with underlying carcinoma). Presents with pruritus, erythema, crusting, ulcers **D**.



Imperforate hymen

Incomplete degeneration of the central portion of the hymen. Accumulation of vaginal mucus at birth → self-resolving bulge in introitus. If untreated, leads to 1° amenorrhea, cyclic abdominal pain, hematocolpos (accumulation of menstrual blood in vagina → bulging and bluish hymenal membrane).

Vaginal tumors**Squamous cell carcinoma**

Usually 2° to cervical SCC; 1° vaginal carcinoma rare.

Clear cell adenocarcinoma

Arises from vaginal adenosis (persistence of glandular columnar epithelium in proximal vagina), found in females who had exposure to diethylstilbestrol in utero.

Sarcoma botryoides

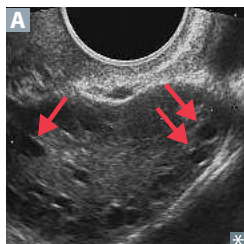
Embryonal rhabdomyosarcoma variant. Affects females < 4 years old; spindle-shaped cells; desmin ⊕. Presents with clear, grapelike, polypoid mass emerging from vagina.

Anovulatory infertility

Infertility 2° to lack of ovulation. Causes:

- PCOS (most common)
- Primary ovarian insufficiency
- Hypogonadotropic hypogonadism
- Hyperprolactinemia

Anovulation is normal during pregnancy, breastfeeding, and near menarche or menopause.

Polycystic ovary syndrome

Unknown cause; associated with dysregulation of ovarian steroidogenesis. ↑ LH:FSH, ↑ androgens (eg, testosterone) from theca interna cells, ↓ rate of follicular maturation → unruptured follicles (cysts) + anovulation. Common cause of ↓ fertility in females.

Diagnosed based on ≥ 2 of the following: cystic/enlarged ovaries on ultrasound (arrows in **A**), oligo-/anovulation, hyperandrogenism (eg, hirsutism, acne). Associated with obesity, insulin resistance, acanthosis nigricans. ↑ risk of endometrial cancer 2° to unopposed estrogen from repeated anovulatory cycles.

Treatment: cycle regulation via weight reduction (↓ peripheral estrone formation), OCPs (prevent endometrial hyperplasia due to unopposed estrogen); ovulation induction for infertility; spironolactone, finasteride, flutamide to treat hirsutism.

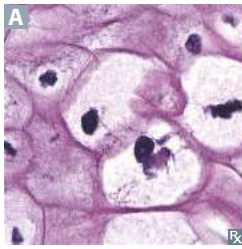
Primary ovarian insufficiency

Also called premature ovarian failure.

Premature atresia of ovarian follicles in females of reproductive age. Most often idiopathic; associated with chromosomal abnormalities (eg, Turner syndrome, fragile X syndrome premutation), autoimmunity. Need karyotype screening. Patients present with signs of menopause after puberty but before age 40. ↓ estrogen, ↑ LH, ↑ FSH.

Functional hypothalamic amenorrhea

Also called exercise-induced amenorrhea. Severe caloric restriction, ↑ energy expenditure, and/or stress → functional disruption of pulsatile GnRH secretion → ↓ LH, FSH, estrogen. Pathogenesis includes ↓ leptin (due to ↓ fat) and ↑ cortisol (stress, excessive exercise). Associated with eating disorders and “female athlete triad” (↓ calorie availability/excessive exercise, ↓ bone mineral density, menstrual dysfunction).

Cervical pathology**Dysplasia and carcinoma in situ**

Disordered squamous epithelial growth; begins at basal layer of squamocolumnar junction (transformation zone) and extends outward. Classified as CIN 1, CIN 2, or CIN 3 (severe dysplasia or carcinoma in situ is unlikely to return to normal), depending on extent of dysplasia. Associated with HPV-16 and HPV-18, which produce both the E6 gene product (inhibits *TP53*) and E7 gene product (inhibits *pRb*) (6 before 7; P before R). Koilocytes (cells with wrinkled “raisinoid” nucleus and perinuclear halo **A**) are pathognomonic of HPV infection. May progress slowly to invasive carcinoma if left untreated. Typically asymptomatic (detected with Pap smear) or presents as abnormal vaginal bleeding (often postcoital).

Risk factors: multiple sexual partners, HPV, smoking, early coitarche, DES exposure, immunocompromise (eg, HIV, transplant).

Invasive carcinoma

Often squamous cell carcinoma. Pap smear can detect cervical dysplasia before it progresses to invasive carcinoma. Diagnose via colposcopy and biopsy. Lateral invasion can block ureters → hydronephrosis → renal failure.

Primary dysmenorrhea

Painful menses, caused by uterine contractions to ↓ blood loss → ischemic pain. Mediated by prostaglandins. Treatment: NSAIDs, acetaminophen, hormonal contraceptives.

Ovarian cysts

Usually asymptomatic, but may rupture, become hemorrhagic, or lead to adnexal torsion.

Follicular cyst

Functional (physiologic) cyst. Most common ovarian mass in young females. Caused by failure of mature follicle to rupture and ovulate. May produce excess estrogen. Usually resolves spontaneously.

Corpus luteal cyst

Functional cyst. Caused by failure of corpus luteum to involute after ovulation. May produce excess progesterone. Usually resolves spontaneously.

Theca lutein cyst

Also called hyperreactio luteinalis. Caused by hCG overstimulation. Often bilateral/multiple. Associated with gestational trophoblastic disease (eg, hydatidiform mole, choriocarcinoma).

Ovarian tumors

Most common adnexal mass in females > 55 years old. Present with abdominal distention, bowel obstruction, pleural effusion.

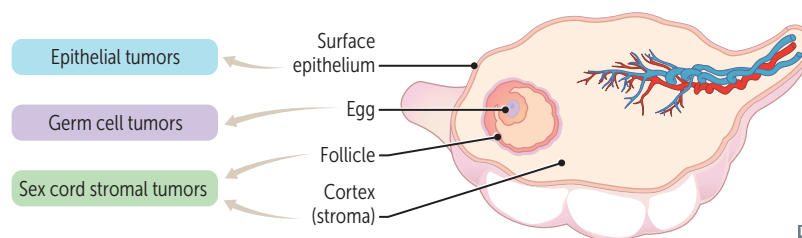
Risk ↑ with advanced age, ↑ number of lifetime ovulations (early menarche, late menopause, nulliparity), endometriosis, genetic predisposition (eg, *BRCA1/BRCA2* mutations, Lynch syndrome).

Risk ↓ with previous pregnancy, history of breastfeeding, OCPs, tubal ligation.

Epithelial tumors are typically serous (lined by serous epithelium natively found in fallopian tubes, and often bilateral) or mucinous (lined by mucinous epithelium natively found in cervix). Monitor response to therapy/relapse by measuring CA 125 levels (not good for screening).

Germ cell tumors can differentiate into somatic structures (eg, teratomas), or extra-embryonic structures (eg, yolk sac tumors), or can remain undifferentiated (eg, dysgerminoma).

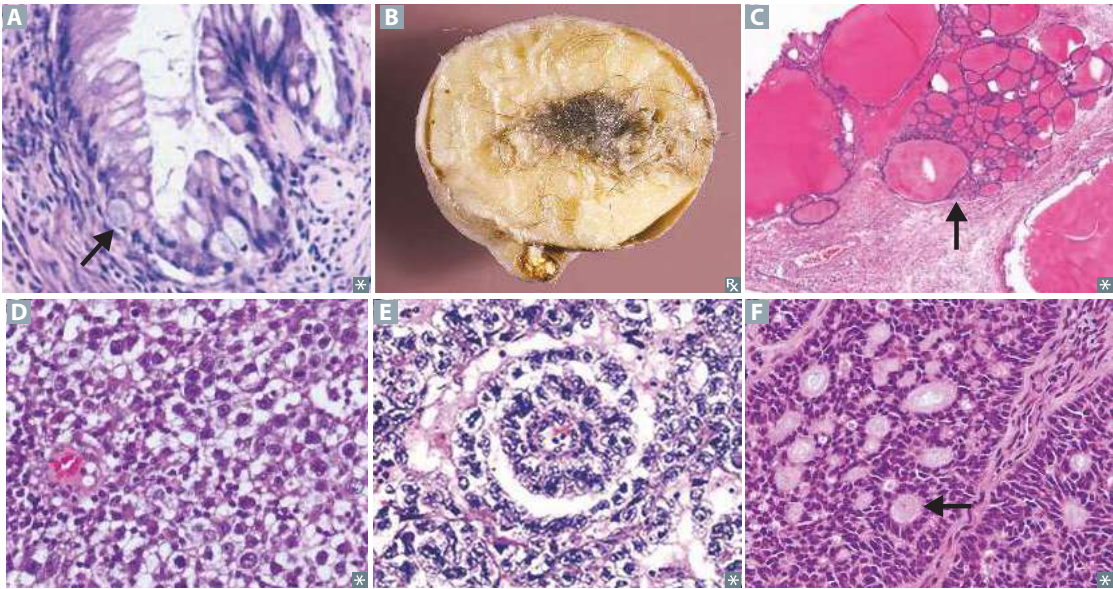
Sex cord stromal tumors develop from embryonic sex cord (develops into theca and granulosa cells of follicle, Sertoli and Leydig cells of seminiferous tubules) and stromal (ovarian cortex) derivatives.



TYPE	CHARACTERISTICS
Epithelial tumors	
Serous cystadenoma	Benign. Most common ovarian neoplasm. Lined by fallopian tube–like epithelium.
Mucinous cystadenoma	Benign. Multiloculated, large. Lined by mucus-secreting epithelium A .
Brenner tumor	Usually benign. Nests of urothelial-like (bladderlike) epithelium with “coffee bean” nuclei.
Serous carcinoma	Most common malignant ovarian neoplasm. Psammoma bodies.
Mucinous carcinoma	Malignant. Rare. May be metastatic from appendiceal or other GI tumors. Can result in pseudomyxoma peritonei (intraperitoneal accumulation of mucinous material, also known as omental caking).
Germ cell tumors	
Mature cystic teratoma	Also called dermoid cyst. Benign. Most common ovarian tumor in young females. Cystic mass with elements from all 3 germ layers (eg, teeth, hair, sebum) B . May be painful 2° to ovarian enlargement or torsion. Monodermal form with thyroid tissue (struma ovarii C) may present with hyperthyroidism. Malignant transformation rare (usually to squamous cell carcinoma).
Immature teratoma	Malignant, aggressive. Contains fetal tissue, neuroectoderm. Commonly diagnosed before age 20. Typically represented by immature/embryoniclike neural tissue.
Dysgerminoma	Malignant. Most common in adolescents. Equivalent to male seminoma but rarer. Sheets of uniform “fried egg” cells D . Tumor markers: ↑ hCG, ↑ LDH.
Yolk sac tumor	Also called endodermal sinus tumor. Malignant, aggressive. Yellow, friable (hemorrhagic) mass. 50% have Schiller-Duval bodies (resemble glomeruli E). Tumor marker: ↑ AFP. Occurs in children and young adult females.

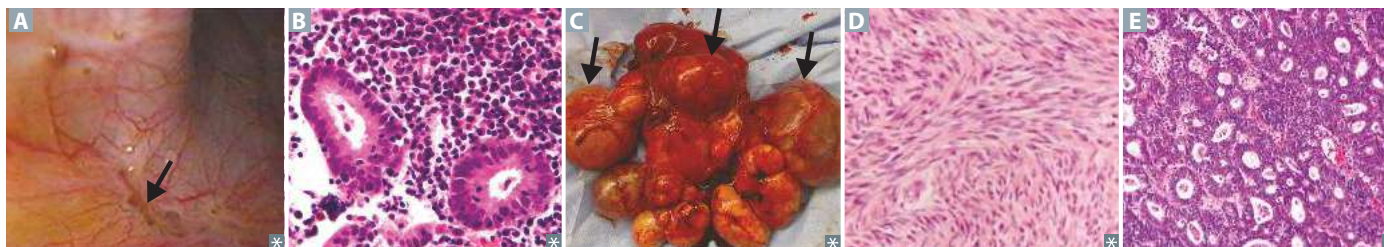
Ovarian tumors (continued)

TYPE	CHARACTERISTICS
Sex cord stromal tumors	
Fibroma	Benign. Bundle of spindle-shaped fibroblasts. Meigs syndrome —triad of ovarian fibroma, ascites, pleural effusion. “Pulling” sensation in groin.
Thecoma	Mostly benign; may produce estrogen. Usually presents as abnormal uterine bleeding in a postmenopausal female.
Sertoli-Leydig cell tumor	Benign. Gray to yellow-brown mass. Resembles testicular histology with seminiferous tubules lined by Sertoli cells and surrounded by interstitial Leydig cells. May produce androgens → virilization (eg, hirsutism, androgenetic alopecia, clitoral enlargement).
Granulosa cell tumor	Most common malignant sex cord stromal tumor. Predominantly occurs in females in their 50s. Often produces estrogen and/or progesterone. Presents with postmenopausal bleeding, endometrial hyperplasia, sexual precocity (in preadolescents), breast tenderness. Histology shows Call-Exner bodies (granulosa cells arranged haphazardly around collections of eosinophilic fluid, resembling primordial follicles; arrow in F). Tumor marker: ↑ inhibin. “Give Granny a Call .”

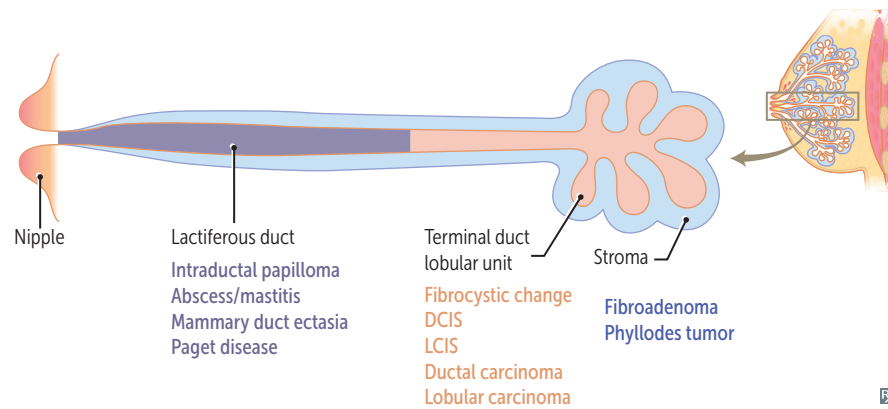


Uterine conditions

TYPE	CHARACTERISTICS
Non-neoplastic	
Adenomyosis	Presence of endometrial tissue (glands and stroma) in myometrium. May be due to invagination of basal layer of endometrium or metaplasia of remnant progenitor cells. Presents with abnormal uterine bleeding, dysmenorrhea. Diffusely enlarged (“globular”), soft (“boggy”) uterus on exam.
Endometriosis	Presence of endometrial tissue outside of the uterus. May be due to ectopic implantation (via retrograde menses, lymphatics, or blood vessels) or metaplasia of remnant progenitor cells. Presents with dysmenorrhea, dyspareunia, infertility, abnormal uterine bleeding. Physical exam is usually normal. However, findings may include immobile uterus, adnexal mass, uterosacral tenderness, and/or nodularity, but are non-diagnostic. Typically involves pelvic sites, such as superficial peritoneum (yellow-brown “powder burn” lesions A) and ovaries (forms blood-filled “chocolate” cysts called endometrioma).
Endometrial hyperplasia	Abnormal endometrial gland proliferation. Usually caused by excess estrogen unopposed by progesterone. Associated with obesity, chronic anovulation (eg, PCOS), nulliparity, HRT, tamoxifen. Presents with abnormal uterine bleeding. ↑ risk for endometrial carcinoma (especially with nuclear atypia).
Endometritis	Inflammation of endometrium B . Usually occurs after delivery due to inoculation of uterine cavity by vaginal microbiota. C-section is the most important risk factor (sutures and necrotic tissue act as nidus for polymicrobial infection). Presents with fever, uterine tenderness, purulent lochia.
Intrauterine adhesions	Fibrous bands/tissue within endometrial cavity. Caused by damage to basal layer of endometrium, usually after dilation and curettage. Presents with abnormal uterine bleeding (↓ menses), infertility, recurrent pregnancy loss, dysmenorrhea. Also called Asherman syndrome when symptomatic.
Neoplastic	
Leiomyoma	Benign tumor of myometrium (also called fibroid). Most common gynecological tumor. Arises in reproductive-age females. ↑ incidence in Black population. Typically multiple; subtypes based on location: submucosal, intramural, or subserosal. Usually asymptomatic, but may present with abnormal uterine bleeding, pelvic pressure/pain, reproductive dysfunction. Estrogen sensitive; tumor size ↑ with pregnancy and ↓ with menopause. Enlarged uterus with nodular contour on exam C . Histology: whorled pattern of smooth muscle bundles D and well-demarcated borders.
Endometrial carcinoma	Malignant tumor of endometrium. Most common gynecological cancer in resource-rich countries. Usually arises in postmenopausal females. Presents with abnormal uterine bleeding. Endometrioid carcinoma —most common subtype of endometrial carcinoma. Associated with long-term exposure to unopposed estrogen. Histology: confluent endometrial glands without intervening stroma E .



Breast pathology



Benign breast diseases

Fibrocystic changes

Most common in premenopausal females 20–50 years old. Present with premenstrual breast pain or lumps; often bilateral and multifocal. Nonproliferative lesions include simple cysts (fluid-filled duct dilation, blue dome), papillary apocrine change/metaplasia, stromal fibrosis. Risk of cancer is usually not increased. Proliferative lesions include

- **Sclerosing adenosis**—acini and stromal fibrosis, associated with calcifications. Slight ↑ risk for cancer.
- **Epithelial hyperplasia**—cells in terminal ductal or lobular epithelium. ↑ risk of carcinoma with atypical cells.

Inflammatory processes

Fat necrosis—benign, usually painless, lump due to injury to breast tissue. Calcified oil cyst on mammography; necrotic fat and giant cells on biopsy. Up to 50% of patients may not report trauma.

Lactational mastitis—occurs during breastfeeding, ↑ risk of bacterial infection through cracks in nipple. *S aureus* is most common pathogen. Treat with antibiotics and continue breastfeeding.

Mammary duct ectasia—dilation of subareolar ducts with inflammation and fibrosis. Associated with smoking. Presents with areolar pain, multicolored discharge, inverted nipple, periareolar mass.

Benign tumors



Fibroadenoma—most common in females < 35 years old. Small, well-defined, mobile mass.

Tumor composed of fibrous tissue and glands. ↑ size and tenderness with ↑ estrogen (eg, pregnancy, prior to menstruation). Risk of cancer is usually not increased.

Intraductal papilloma—small fibroepithelial tumor within lactiferous ducts, typically beneath areola.

Most common cause of nipple discharge (often unilateral, serous or bloody). Slight ↑ risk for cancer.

Phyllodes tumor—large mass of connective tissue and cysts with “leaflike” lobulations **A**. Most common in 5th decade. Some may become malignant.

Gynecomastia

Breast enlargement in males due to ↑ estrogen compared with androgen activity. Physiologic in newborn, pubertal, and older males, but may persist after puberty. Other causes include cirrhosis, hypogonadism (eg, Klinefelter syndrome), testicular tumors, drugs (eg, spironolactone).

Breast cancer

Commonly postmenopausal. Often presents as a palpable hard mass **A** most often in upper outer quadrant. Invasive cancer can become fixed to pectoral muscles, deep fascia, Cooper ligaments, and overlying skin → nipple retraction/skin dimpling.

Usually arises from terminal duct lobular unit. Amplification/overexpression of estrogen/progesterone receptors or HER2 (an EGF receptor) is common; triple negative (ER $\ominus$, PR $\ominus$, and HER2 $\ominus$) form more aggressive.

Risk factors in females: $\uparrow$ age; history of atypical hyperplasia; family history of breast cancer; race (White patients at highest risk, Black patients at $\uparrow$ risk for triple $\ominus$ breast cancer); *BRCA1/BRCA2* mutations; $\uparrow$ estrogen exposure (eg, nulliparity); postmenopausal obesity (adipose tissue converts androstenedione to estrone); $\uparrow$ total number of menstrual cycles; absence of breastfeeding; later age of first pregnancy; alcohol intake. In males: *BRCA2* mutation, Klinefelter syndrome. Axillary lymph node metastasis most important prognostic factor in early-stage disease.

TYPE	CHARACTERISTICS	NOTES
Noninvasive carcinomas		
Ductal carcinoma in situ	Fills ductal lumen (black arrow in B indicates neoplastic cells in duct; blue arrow shows engorged blood vessel). Arises from ductal atypia. Often seen early as microcalcifications on mammography.	Early malignancy without basement membrane penetration. Usually does not produce a mass.
Paget disease	Extension of underlying DCIS/invasive breast cancer up the lactiferous ducts and into the contiguous skin of nipple → eczematous patches over nipple and areolar skin C .	Paget cells = intraepithelial adenocarcinoma cells.
Lobular carcinoma in situ	$\downarrow$ E-cadherin expression. No mass or calcifications → incidental biopsy finding.	$\uparrow$ risk of cancer in either breast (vs DCIS, same breast and quadrant).
Invasive carcinomas		
Invasive ductal	Firm, fibrous, “rock-hard” mass with sharp margins and small, glandular, ductlike cells in desmoplastic stroma.	Most common type of invasive breast cancer.
Invasive lobular	$\downarrow$ E-cadherin expression → orderly row of cells (“single file” D) and no duct formation. Often lacks desmoplastic response.	Often bilateral with multiple lesions in the same location. Lobular carcinoma lacks cadherin and forms lines of cells.
Inflammatory	Dermal lymphatic space invasion → breast pain with warm, swollen, erythematous skin around exaggerated hair follicles (peau d’orange) E .	Poor prognosis (50% survival at 5 years). Often mistaken for mastitis or Paget disease. Usually lacks a palpable mass.



Penile pathology**Peyronie disease**

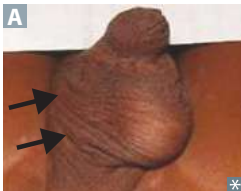
Abnormal curvature of penis **A** due to fibrous plaque within tunica albuginea. Associated with repeated minor trauma during intercourse. Can cause pain, anxiety, erectile dysfunction. Consider surgical repair or treatment with collagenase injections once curvature stabilizes. Distinct from penile fracture (rupture of tunica albuginea due to forced bending).

Ischemic priapism

Painful sustained erection lasting > 4 hours. Associated with sickle cell disease (sickled RBCs block venous drainage of corpus cavernosum vascular channels), medications (eg, sildenafil, trazodone). Treat immediately with corporal aspiration, intracavernosal phenylephrine, or surgical decompression to prevent ischemia.

Squamous cell carcinoma

Seen in the US, but more common in Asia, Africa, South America. Most common type of penile cancer **B**. Precursor in situ lesions: Bowen disease (in penile shaft, presents as leukoplakia “white plaque”), erythroplasia of Quey^{rat} (carcinoma in situ of the glans, presents as erythroplakia “red plaque”), Bowenoid papulosis (carcinoma in situ of unclear malignant potential, presenting as reddish papules). Associated with uncircumcised males and HPV-16.

Cryptorchidism

Descent failure of one **A** or both testes. Impaired spermatogenesis (since sperm develop best at temperatures < 37°C) → subfertility. Can have normal testosterone levels (Leydig cells are mostly unaffected by temperature). Associated with ↑ risk of germ cell tumors. Prematurity ↑ risk of cryptorchidism. ↓ inhibin B, ↑ FSH, ↑ LH; testosterone ↓ in bilateral cryptorchidism, normal in unilateral. Most cases resolve spontaneously; otherwise, orchiopexy performed before 2 years of age.

Testicular torsion

Rotation of testicle around spermatic cord and vascular pedicle. Commonly presents in males 12–18 years old. Associated with congenital inadequate fixation of testis to tunica vaginalis → horizontal positioning of testes (“bell clapper” deformity). May occur after an inciting event (eg, trauma) or spontaneously. Characterized by acute, severe pain, high-riding testis, and absent cremasteric reflex. ⊖ Prehn sign.

Treatment: surgical correction (orchiopexy) within 6 hours, manual detorsion if surgical option unavailable in timeframe. If testis is not viable, orchiectomy. Orchiopexy, when performed, should be bilateral because the contralateral testis is at risk for subsequent torsion.

Varicocele

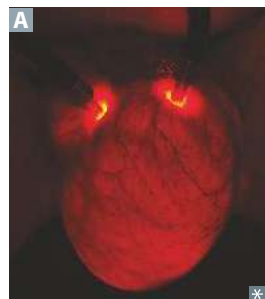
Dilated veins in pampiniform plexus due to ↑ venous pressure; most common cause of scrotal enlargement in adult males. Most often on left side because of ↑ resistance to flow from left gonadal vein drainage into left renal vein. Right-sided varicocele may indicate IVC obstruction (eg, from RCC invading right renal vein). Can cause infertility because of ↑ temperature. Diagnosed by standing clinical exam/Valsalva maneuver (distension on inspection and “bag of worms” on palpation; augmented by Valsalva) or ultrasound **A**. Does not transilluminate. Treatment: consider surgical ligation or embolization if associated with pain or infertility.

Extragenadal germ cell tumors

Arise in midline locations. In adults, most commonly in retroperitoneum, mediastinum, pineal, and suprasellar regions. In infants and young children, sacrococcygeal teratomas are most common.

Benign scrotal lesions

Testicular masses that can be transilluminated **A** (vs solid testicular tumors).

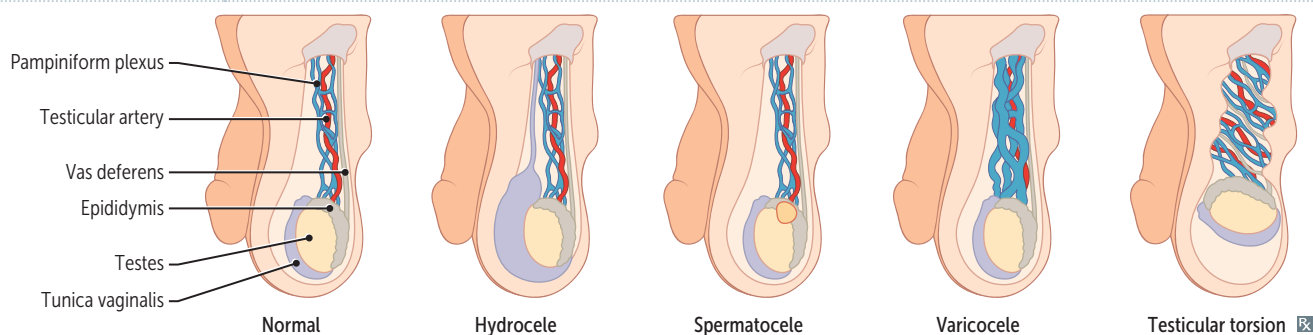
Hydrocele

Accumulation of serous fluid within tunica vaginalis. Types:

- **Congenital** (communicating)—due to incomplete obliteration of processus vaginalis. Common cause of scrotal swelling in infants. Most resolve spontaneously within 1 year.
- **Acquired** (noncommunicating)—due to infection, trauma, tumor. Termed hematocele if bloody.

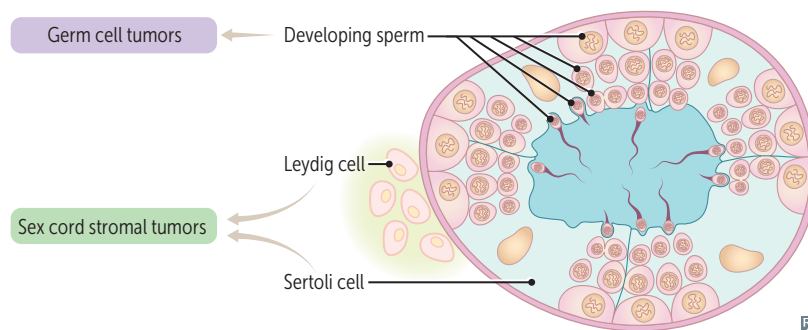
Spermatocele

Cyst due to dilated epididymal duct or rete testis. Paratesticular fluctuant nodule on palpation.

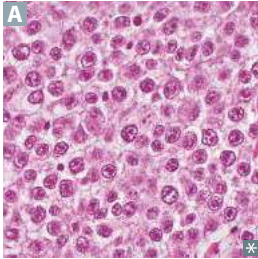
**Testicular tumors**

Germ cell tumors account for ~ 95% of all testicular tumors. Arise from germ cells that produce sperm. Most often occur in young males. Risk factors: cryptorchidism, Klinefelter syndrome. Can present as mixed germ cell tumors. Do not transilluminate. Usually not biopsied (risk of seeding scrotum), removed via radical orchiectomy.

Sex cord stromal tumors develop from embryonic sex cord (develops into Sertoli and Leydig cells of seminiferous tubules, theca and granulosa cells of follicle) derivatives. 5% of all testicular tumors. Mostly benign.



Testicular tumors (continued)

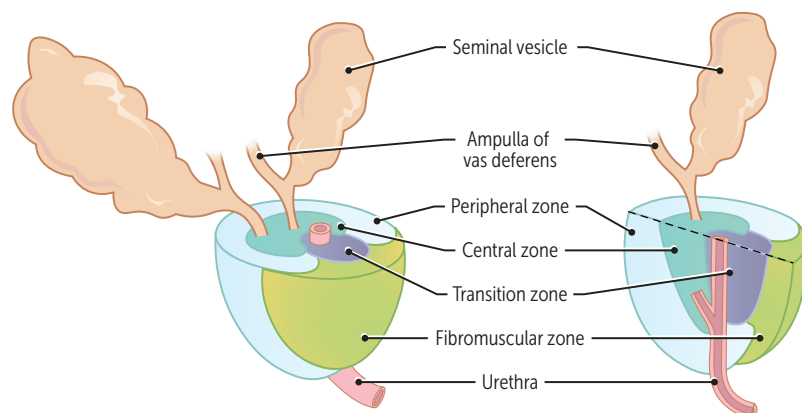
TYPE	CHARACTERISTICS
Germ cell tumors	
Seminoma	Malignant. Painless, homogenous testicular enlargement. Most common testicular tumor. Analogous to ovarian dysgerminoma. Does not occur in infancy. Large cells in lobules with watery cytoplasm and “fried egg” appearance on histology A , ↑ placental alkaline phosphatase (PLAP). Highly radiosensitive. Late metastasis, excellent prognosis.
	
Embryonal carcinoma	Malignant. Painful, hemorrhagic mass with necrosis. “Pure” embryonal carcinoma is rare; most commonly mixed with other tumor types. May present with metastases. May be associated with ↑ hCG and normal AFP levels when pure (↑ AFP when mixed). Worse prognosis than seminoma.
Teratoma	Mature teratoma may be malignant in adult males. Benign in children and females.
Yolk sac tumor	Also called endodermal sinus tumor. Malignant, aggressive. Yellow, mucinous. Analogous to ovarian yolk sac tumor. Schiller-Duval bodies resemble primitive glomeruli. ↑ AFP is highly characteristic. Most common testicular tumor in children < 3 years old.
Choriocarcinoma	Malignant. Disordered syncytiotrophoblastic and cytotrophoblastic elements. Hematogenous metastases to lungs and brain. ↑ hCG. May produce gynecomastia, symptoms of hyperthyroidism (hCG and TSH share an identical α subunit and a similar β subunit, which determines their hormonal function).
Non-germ cell tumors	
Leydig cell tumor	Mostly benign. Golden brown color; contains Reinke crystals (eosinophilic cytoplasmic inclusions). Produces androgens or estrogens → precocious puberty, gynecomastia.
Sertoli cell tumor	Also called androblastoma (arises from sex cord stroma). Mostly benign.
Primary testicular lymphoma	Malignant, aggressive. Typically diffuse large B-cell lymphoma. Often bilateral. Most common testicular cancer in males > 60 years old.

Hormone levels in germ cell tumors

	SEMINOMA	YOLK SAC TUMOR	CHORIOCARCINOMA	TERATOMA	EMBRYONAL CARCINOMA
PLAP	↑	—	—	—	—
AFP	—	↑↑	—	—/↑	—/↑ (when mixed)
β-hCG	—/↑	—/↑	↑↑	—	↑

Epididymitis and orchitis	Most common causes: ▪ <i>C trachomatis</i> and <i>N gonorrhoeae</i> (young males) ▪ <i>E coli</i> and <i>Pseudomonas</i> (older males, associated with UTI and BPH) ▪ Autoimmune (eg, granulomas involving seminiferous tubules)
Epididymitis	Inflammation of epididymis. Presents with localized pain and tenderness over posterior testis. ⊕ Prehn sign (pain relief with scrotal elevation). May progress to involve testis.
Orchitis	Inflammation of testis. Presents with testicular pain and swelling. Mumps orchitis ↑ infertility risk. Rare in males < 10 years old.

Benign prostatic hyperplasia



Common in males > 50 years old. Characterized by smooth, elastic, firm nodular enlargement (hyperplasia not hypertrophy) of transition zone, which compress the urethra into a vertical slit. Not premalignant.

Often presents with ↑ frequency of urination, nocturia, difficulty starting and stopping urine stream, dysuria. May lead to distention and hypertrophy of bladder, hydronephrosis, UTIs. ↑ total PSA, with ↑ fraction of free PSA. PSA is made by prostatic epithelium stimulated by androgens.

Treatment: α_1 -antagonists (terazosin, tamsulosin), which cause relaxation of smooth muscle; 5 α -reductase inhibitors (eg, finasteride); PDE-5 inhibitors (eg, tadalafil); surgical resection (eg, TURP, ablation).

Prostatitis

Characterized by dysuria, frequency, urgency, low back pain. Warm, tender, enlarged prostate. Acute bacterial prostatitis—in older males most common bacterium is *E coli*; in young males consider *C trachomatis*, *N gonorrhoeae*.

Chronic prostatitis—either bacterial or nonbacterial (eg, 2° to previous infection, nerve problems, chemical irritation).

Prostatic adenocarcinoma

Common in males > 50 years old. Arises most often from posterior lobe (peripheral zone) of prostate gland and is most frequently diagnosed by ↑ PSA and subsequent needle core biopsies (transrectal, ultrasound-guided). Histologically graded using Gleason grade, which is based on glandular architecture and correlates closely with metastatic potential. Prostatic acid phosphatase (PAP) and PSA are useful tumor markers (↑ total PSA, with ↓ fraction of free PSA). Osteoblastic metastases in bone may develop in late stages, as indicated by lower back pain and ↑ serum ALP and PSA. Metastasis to the spine often occurs via Batson (vertebral) venous plexus.